RETAIL SALES ANALYSIS

Excel Project | README & Project Documentation

A practical Excel-based analysis of retail transactions to understand sales, profit, product performance, regional contribution and monthly trends.

ORDERS / RECORDS	TOTAL SALES	TOTAL PROFIT	UNITS SOLD
559	■17,049,046	■319,886	1,652

1. Project objective

The goal of this project is to turn raw retail sales records into useful business information. The analysis focuses on where sales are coming from, which product categories and products contribute most, how profit varies across regions and categories, and where loss-making transactions may need attention.

2. Dataset overview

Source: Retail_Sales worksheet in the supplied Excel workbook
Records: 559 transaction rows (excluding the header)
Columns: 11
Period: 01 Jan 2024 to 01 Jul 2024
Fields: Order_ID, Order_Date, Region, State, Category, Sub_Category, Product_Name, Sales, Quantity, Profit and Customer_Segment.

3. Workbook structure

Worksheet	Purpose
Retail_Sales	Original transaction-level retail sales data.
Data_Preparation	Working sheet for preparing and organizing the dataset.
Data_Analysis	Analysis view with sales and profit summaries by region and category.
Pivot_Table	Pivot-style summaries for region, category, product and month.
Data_Visualization	Contains three Excel charts for visual exploration.

4. Analysis workflow

• **1. Inspect the source data:** review column names, data types, date range and transaction fields.

• **2. Prepare the data:** use the preparation worksheet as the working copy and check for missing values, duplicate rows and unusual profit values.

• **3. Analyze performance:** summarize sales, quantity and profit by region, category, product and month.

• **4. Build summaries:** use pivot-style tables to compare contributions and identify leading segments.

• **5. Visualize results:** use the charts in the visualization worksheet to make comparisons easier to interpret.

5. Key findings from the supplied data

Sales by region

Region	Sales	Share of total
North	■7,041,263	41.30%
South	■3,425,569	20.09%
East	■3,354,621	19.68%
West	■3,227,593	18.93%

Sales and profit by category

Category	Sales	Profit	Sales share
Technology	■6,911,923	■139,146	40.54%
Furniture	■6,909,530	■125,174	40.53%
Office Supplies	■3,227,593	■55,566	18.93%

• The **North** region has the highest sales at ■7,041,263 (41.30% of total sales).

• **Technology** has the highest sales at ■6,911,923; **Technology** generates the highest category profit at ■139,146.

• **Office Chair** is the top-selling product by sales value at ■3,554,909.

• 129 transactions have negative profit. These records are worth reviewing for discounting, cost, pricing or data-quality issues.

• Total sales are shown as ■17,049,046 and total profit as ■319,886 for the records in this workbook.

6. Data quality checks

The source sheet contains 0 blank cells across the 11 fields and 0 fully duplicated rows. Negative profit values are present in 129 records; these have not been treated as errors automatically because a negative profit can be a valid business outcome.

7. Tools and Excel concepts

Microsoft Excel • structured tabular data • sorting and filtering • data validation and quality checks • aggregation and comparison • PivotTables / pivot-style summaries • charts • percentage contribution analysis.

8. How to use this project

1. Open the Excel workbook and begin with **Retail_Sales** to understand the source fields.
2. Review **Data_Preparation** and **Data_Analysis** to follow the preparation and analysis layout.

3. Explore **Pivot_Table** to compare region, category, product and month-level results.
4. Open **Data_Visualization** to review the three charts.
5. Change filters or refresh PivotTables if you add or update source records, then recheck the totals.

9. Project summary

This project demonstrates how Excel can be used to organize retail transaction data, calculate and compare business metrics, identify top-performing segments and highlight negative-profit transactions. It is suitable as an entry-level Data Analyst / MIS portfolio project.

Note: All figures in this README are calculated from the supplied workbook's Retail_Sales sheet. Currency is presented in Indian rupees (■) for readability; confirm the source dataset's intended currency before using the figures in a business report.